Hybrid EfficientNetV2-B1 and Multi-Head Self Attention Transformer for Fish and Shrimp Disease Classification

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***Abstract*—Fish disease is a major problem in the global market causing estimated losses of billions of US dollars. Developing countries suffer more production and economic hits. Diseased fish become harder to sell, causing market value and reputation loss for farmers. To address the issue, we looked for a hybrid ensemble approach. We used EfficientNetV2-B1 for its efficient feature extraction performance and paired it up with a multi-head self-attention transformer for overall spatial context of fish anatomy. Evaluated on the BD Fish & Shrimp dataset comprising 5,887 images, we employed a soft voting mechanism between the two models to hit a state-of-the-art 96.30% accuracy. These results provide high confidence in predicting 11 classes of diseases across fish and shrimp yielding zero false positives for healthy shrimp and a near-zero false positive rate for healthy fish. This significantly guides the farmers in assessing the conditions of their livestock, enabling timely management and actions.**

***Index Terms*—aquaculture, deep learning, EfficientNetV2, vision transformer, disease classification, ensemble learning**

I. Introduction

Aquatic farming is a rapidly growing industry, with a valuation of approximately USD 322 billion in 2024, dominated by the Asia-Pacific region [1]. With innovation in the technologies, improved logistics, increasing consumer expenditures, and a preference for fast foods, the demand for sea products has increased rapidly. The rising demand puts pressure on timely and quality supply. However, fish are highly prone to various fungal, bacterial, and parasitic diseases. These diseases hamper their growth, degrade the product quality, and increase mortality rates. Finally, this leads to expensive treatments, causing revenue deficits, degradation of product quality and damaging consumer confidence [2].

Typically, farmers tackle this using bare visual cues, which is time-consuming and inaccurate. This affects further management. Furthermore, it is difficult to distinguish healthy fish from diseased ones due to factors like low visibility, poor water quality, and high turbidity. Also, natural anomalies like birthmarks and pigmentations can cause false positives, all interrupting the treatment plans.

In current studies, researchers use deep learning, like Convolutional Neural Networks (CNNs), for aquatic disease detection. CNNs build detailed feature maps of the subject of study by sliding across each pixel [3]. But it has been proven that these neural networks have a vanishing gradient problem. Residual Networks (ResNet) came to the rescue to solve this issue by allowing skip connections to go in deeper layers for

better feature extraction [4]. Still, these models focused mostly on the specific pixels of the specific specimen for creating and learning patterns on smaller targeted spaces, leaving out the overall spatial structure of the specimen's body. For example, it will struggle to distinguish a white dot present on the fin of the fish from a reflection rather than a specific disease; it has no knowledge of where on the body of the fish it is looking.

Now, it is easy to fool these models; factors like water turbidity, dirt, and low image quality can easily confuse these narrow-visioned models, triggering false alarms for the farmers. The latest development in overcoming this flaw is using ensembles of models [5]. The most recent papers suggest using ResNet-50 (25 million parameters) and VGG16 (138 million parameters) [4]. These models do achieve great numbers. But in achieving that, they create huge computational overhead and GPU expenses. Deploying such huge models is not feasible for a rural farmer using a simple mobile device. Therefore, we need a different approach to achieve high accuracy and solve the issue of these big ensemble models' parameters and power consumption.

To achieve high accuracy with less computational overhead, we propose a hybrid approach. We combine two, Models A and B. In Model A, we used the EfficientNetV2-B1 CNN model [6] for efficient pixel focusing and to resolve the global pattern loss we combined it with a Multi-Head Self-Attention transformer [7]. For Model B, we used a pure CNN (EfficientNetV2-B0) to optimize the local textures of the subject. We then used a soft voting mechanism between Model A and B to combine their results probabilistically [5]. This approach gave us an accuracy of 96.30% across a complex 11-class dataset of both fish and shrimp. Importantly, this ensemble showed zero false positives in predicting healthy shrimp, and a near-zero false positive rate for healthy fish, minimizing the risk of untreated livestock.

II. Related Work

Single deep learning architectures have severe limitations in underwater conditions, as previous studies show. For instance, MobileNet-V2 and Inception-V3 models were evaluated against fish datasets, yielding only 68.09% and 64.09% accuracy, respectively. To improve this performance, Biswas et al. proposed a feature fusion method [3]. The extracted feature vectors from VGG-16, MobileNet-V2, and Inception-V3 are then passed to a Support Vector Machine (SVM) rather than typical classification layers. This fusion method achieved an accuracy of 96.88%. This shows that model fusion is a more optimal method than single deep learning architecture for underwater conditions.

Another common method is to use weighted ensembles to increase accuracy. Al Maruf et al. [15] developed a model with four backbones, including ResNet-50, Inception-V3, VGG-16, and MobileNet-V2. Performance Metric-Infused Weighted Ensemble (PMIWE) approach was used to calculate the weights using a hyperbolic tangent function based on the F1-scores. This experimentation with PMIWE achieved a 97.53% testing accuracy. The accuracy is great but the simultaneous execution of four heavy pre-trained models is computationally intensive and requires a lot of memory. This makes deploying the system on the low-cost edge devices fish farmers use in practice a huge challenge.

While standard CNN can detect local texture anomalies effectively, it cannot infer spatial changes of a fish's body brought about by diseases. This is addressed by Utku et al. [16] in 2025 through designing a hybrid model that attaches a ResNet-50 CNN to a Vision Transformer (ViT). The CNN extracts the basic visual features and the ViT accounts for the global spatial context at 99.14% accuracy. Transformers, however, are known for requiring high-tier hardware resources. On the contrary, Suleiman et al. [17] concentrated on optimizing model performance using limited hardware for Nigerian aquaculture. Suleiman et al. achieved a 98.5% accuracy by pairing a ResNet-50 with a computationally inexpensive EfficientNet-B0 backbone.

There is an obvious trade-off when looking at the recent papers. Such models that pay attention to the global spatial context (ViTs) are too heavy for mobile deployment. Lightweight models lack the ability to manage complicated spatial features related to reflections on the water's surface. Aquaculture needs a method that can grasp the global anatomical structure like a Transformer but is lightweight enough to run on a mobile phone without the need for GPUs.

III. Proposed Deep Learning Model

In this research study, two different structures of the EfficientNetV2 pre-trained models (EfficientNetV2-B1 and EfficientNetV2-B0) were combined into one model that could identify aquatic disease based on many complex real-life scenarios. These scenarios could be characterized by ponds with lots of sediment and/or very little light. Instead of using a single, large neural network on edge devices, this model is based on a dual-branch heterogeneous architecture to reduce overall computational loads on each device. The combination of these specially optimized, lightweight models using transfer learning will provide benefits in terms of diagnostic accuracy when identifying aquatic disease outbreaks, while operating well within the computation and thermal limits of the common mobile processors used by rural aquaculture farmers. This will ultimately allow for quick and effective responses to identify aquatic disease outbreaks.

In terms of working, the proposed architecture, as illustrated in Fig. 1, concurrently processes the input image tensor through its two parallel branches. Let the input aquatic image be defined as a tensor $X \in \mathbb{R}^{H \times W \times C}$, where the spatial dimensions are $H = 224$ and $W = 224$, with $C = 3$ RGB color channels. This tensor is divided into two branches - a global spatial one and a texture one. The global spatial branch uses the EfficientNetV2-B1 architecture along with an MHSA mechanism to infer both global spatial context and global size anomalies in the anatomy of the specimen. The localized branch performs as a dedicated local feature extractor (acts as a digital microscope) to identify only locally identified pathological anomalies like small numbered colonies of bacteria. This

branch uses an EfficientNetV2-B0 backbone and totally omits transformer logic thereby lowering the computational burden. Once both the global and local feature representations have been derived from the respective branches their associated confidence scores are probabilistically combined using a soft voting ensemble approach resulting in the final diagnostic prediction.

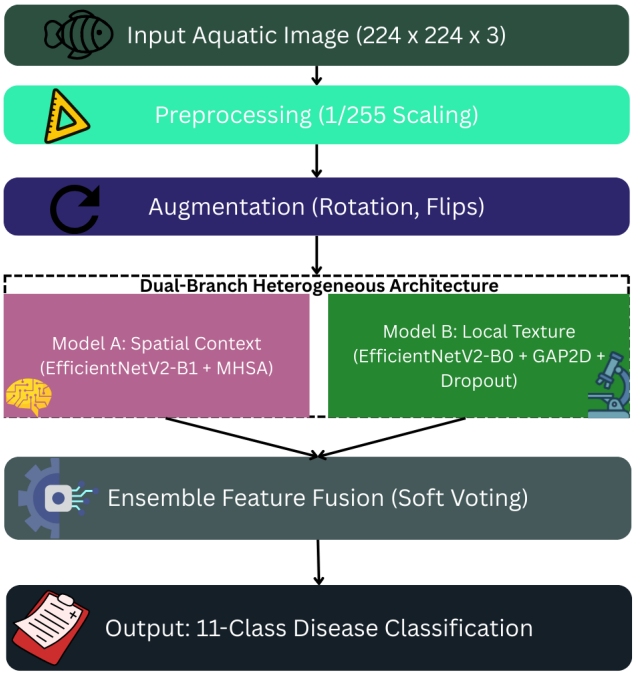


Fig. 1. Proposed Heterogeneous Feature-Fused Ensemble Architecture.

Data Preprocessing and Augmentation

The development of this model utilized the BD Fish & Shrimp dataset [12], available via the Hugging Face open-source platform. The dataset consists of 5,887 high-quality images categorized into 11 different aquatic conditions, encompassing healthy individuals as well as those suffering from bacterial, fungal, or parasitic infections. During the initial preprocessing phase, all images underwent resizing to the dimensions of 224×224 pixels. The images were then normalized by a factor of $1/255$, ensuring that each color channel contained values ranging from 0 to 1. To harden the model against real-world environmental factors [9], we implemented digital augmentation techniques designed to introduce high-variance optical noise. To simulate the results created by non-professional photographers capturing subjects in and around murky water, training images were subjected to random rotations (from -20 degrees to +20 degrees), horizontal flipping, and random variations of 15% in terms of width, height, shear, and zoom. Finally, during these spatial transformations, new boundary pixels were generated using Nearest Neighbor Interpolation to preserve edge continuity.

The Hybrid Ensemble Model

The proposed architecture relies heavily on the EfficientNetV2 backbone to extract baseline feature maps efficiently without excessive computational weight. Specifically, the spatial branch utilizes EfficientNetV2-B1, while the texture branch utilizes EfficientNetV2-B0, acting as a digital magnifier searching specifically for localized anomalies on the specimen, such as small bacterial spots. EfficientNetV2 optimizes standard convolutional layers by uniformly scaling network width (w), depth (d), and resolution (r) using a compound coefficient ϕ, governed by the mathematical constraints:

$$d = \alpha^{\phi}, w = \beta^{\phi}, r = \gamma^{\phi} \quad \text{s.t.} \quad \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \tag{1}$$

This uniform scaling allows both backbones to maximize feature extraction while remaining computationally lightweight.

To create global anatomic context, we combined the EfficientNetV2-B1 with a Multi-Head Self-Attention (MHSA) mechanism. The resulting feature map is flattened out into 49 separate patches/tokens that are processed through 4 heads of attention, with the key dimensions of each head being 1280. Mathematically, the attention function maps queries (Q), keys (K), and values (V) using scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

This process reduces the likelihood of the model misclassifying an environmental feature; i.e. the model learns through attention that the bright white reflection above the dorsal fin is current illumination rather than a fungal infection.

The ensemble's main strength comes from creating two independent branches that cross-validate each other. To ensure both models would learn different feature representations and generate independent errors when making predictions, we developed Model B using a normal Adam optimizer [10], in contrast to Model A which used Adamax. Furthermore, local features from Model B were aggregated by GAP2D and dropped out at 30% to avoid overfitting. The final prediction is calculated by averaging their softmax probability vectors [13]. We use the argmax function to select the winning class from the two models:

$$P_{\text{final}} = \arg\max\left(\frac{P_A + P_B}{2}\right) \tag{3}$$

Therefore, when Model B generates a false positive because of a localized artifact related to texture, Model A's certainty of spatiality will suppress the false positive error and reduce the probability of making an incorrect classification back down to the mean.

IV. Results and Discussion

A. Evaluation Metrics

In addition to overall accuracy of the model we evaluated its performance related to precision, recall and F1-score for each classification using standard calculation techniques. Precision measures how many of the predicted diseased fish (i.e., true positives out of all predicted positives) were actually diseased.

Recall measures whether all diseased fish (i.e. true positives) in the dataset were found. F1-score is the harmonic mean between precision and recall and therefore provides a balanced perspective of performance of the classifications.

B. Classification Metrics and Study of Feature Selection

Although the mathematical definitions of the metrics are observably standard, the ablation data presented in Table I confirms the ability of the architecture to meet its class goals. We performed a comparison on the two branches (the Hybrid Spatial Context Network (Model A) and the Local Texture Specialist (Model B)) to determine the individual contribution of each branch, prior to fusing the features from both branches. The Hybrid Spatial Context Network (Model A) produced a stand-alone accuracy of 94.61% and a macro F1-score of 91.09% when run independently of the Local Texture Specialist (Model B). The Local Texture Specialist (Model B) produced a stand-alone accuracy of 95.45% and a macro F1-score of 92.01%.

However, when considered independently, both models were still susceptible to false positives when local textures were inconsistent with their corresponding global geometry. Fusing the two sets of features via soft voting resolves the inaccuracy from the independently generated models [13]. Following the feature fusion process, the fused classifier produced an aggregate accuracy score of 96.30%. The macro precision value of 93.89% and the macro recall of 93.55% resulted in an overall fused F1-Score of 93.56%. The F1-score metric increase of 1.5%–2.5% in contrast to the respective, independent Spatial and Texture Classifier models provides mathematical validation that cross-validating a Spatial Transformer through a Texture-oriented CNN decreases the count of misclassified instances in aquatically biased datasets with a high degree of imbalance across classes.

TABLE I
ABLATION STUDY OF PROPOSED ARCHITECTURE

Model	Accuracy	Precision	Recall	F1-Score
Model A (EfficientNetV2-B1 + MHSA)	94.61%	91.79%	91.16%	91.09%
Model B (EfficientNetV2-B0)	95.45%	92.23%	92.03%	92.01%
Ensemble Model	**96.30%**	**93.89%**	**93.55%**	**93.56%**

C. Performance Metric and Confusion Matrix

As depicted in Fig. 2, a Confusion Matrix was generated in final testing phase for classifying the classification accuracy (96.30% accuracy of the ensemble) across all 11 classifications.Furthermore, the detailed class-wise precision, recall, and F1-scores are summarized in Table II. A significant amount of diagonal activity was observed in the Confusion Matrix which suggests that the majority of testing data images had been accurately labelled (classified) by the model. It was also shown that the healthy class had no false positive results in the Confusion Matrix. Therefore, this metric is highly beneficial for real-world aquaculture scenarios, ensuring that an aqua farmer is not erroneously advised to apply expensive or hazardous chemicals to a perfectly healthy pond.

TABLE II
CLASS-WISE PERFORMANCE METRICS (PROPOSED MODEL)

Class Acronym	Precision	Recall	F1-Score
FBRD	85.71%	96.77%	90.91%
FBA	90.00%	90.00%	90.00%
FBGD	90.00%	93.10%	91.53%
FFDS	90.00%	90.00%	90.00%
FHF	96.43%	90.00%	93.10%
FPD	82.35%	90.32%	86.15%
FVWTD	100.00%	80.65%	89.29%
SBG	98.25%	100.00%	99.12%
SH	100.00%	100.00%	100.00%
SWSSV	100.00%	98.18%	99.08%
SWSSV-BG	100.00%	100.00%	100.00%

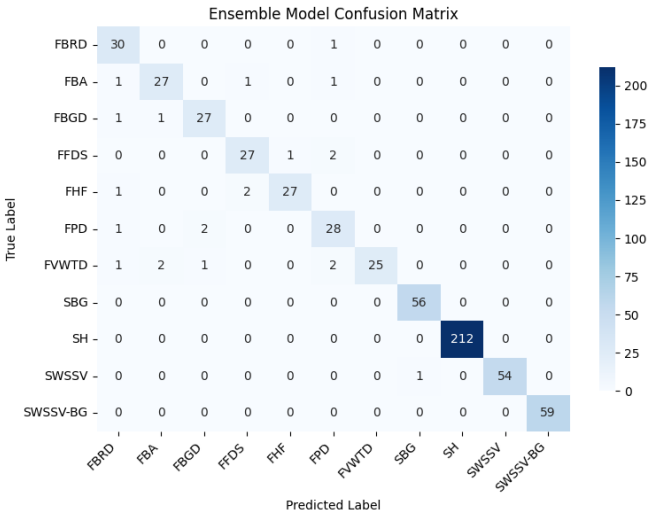


Fig. 2. Confusion matrix of the proposed ensemble model using testing data. Acronyms: FBRD (Fish Bacterial Red Disease), FBA (Fish Bacterial Aeromoniasis), FBGD (Fish Bacterial Gill Disease), FFDS (Fish Fungal Saprolegniasis), FHF (Fish Healthy), FPD (Fish Parasitic Disease), FVWTD (Fish Viral White Tail Disease), SBG (Shrimp Black Gill), SH (Shrimp Healthy), SWSSV (Shrimp White Spot Syndrome Virus), SWSSV-BG (Shrimp White Spot & Black Gill).

D. Hardware Efficiency and Parameter Budget

While it's important to achieve high levels of accuracy, there is no point in having an ML model that doesn't allow for execution on edge devices within rural areas of the world. As a typical legacy architecture, VGG-16 has about 138 million unique weights that create a very large computational burden. The architecture used as part of our redesign has been designed with a focus on eliminating this large computational burden by avoiding the use of large legacy architectures. The pure-CNN EfficientNetV2-B0 [6] (Model B) requires only 7.1 million parameters, while the spatial branch (Model A) has an EfficientNetV2-B1 backbone along with a dense multi-head attention mechanism (with 1,280 key dimensions across 4 heads), resulting in 33.8 million total parameters. When combined, the feature-fused ensemble uses approximately 40.9 million parameters, which is a 70% reduction in computational weight compared to VGG-16 architectures, thereby allowing it to operate on standard mobile processors without excessive

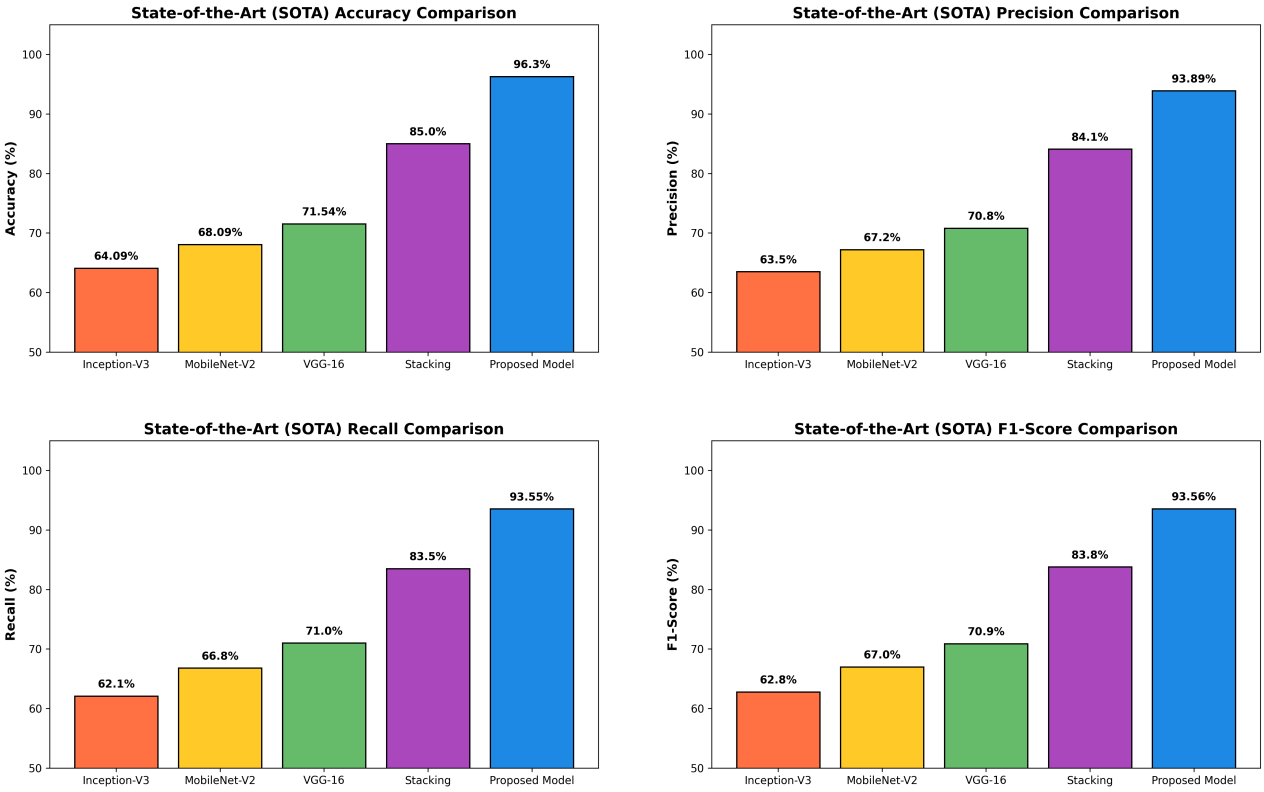


Fig. 3. Performance comparison across multiple metrics between baseline architectures and the proposed model.

thermal throttling during inference.

E. Comparative Assessment Against Market Benchmarks

The efficacy of the new hybrid architecture was evaluated in relation to traditional baseline models. Prior studies have indicated that single models such as Inception-V3 and VGG-16 struggle to process underwater images, achieving only between 64% and 71% accuracy at their peak performance. A single EfficientNet-B0 is capable of significantly better accuracy than either of these models but lacks a comprehensive understanding of the overall anatomical spatial context [6]. As shown in Fig. 3, our proposed soft-vote ensemble demonstrates significant performance improvements across all four key evaluation metrics when compared to the baseline single-architecture models. While baseline models like VGG-16 and Stacking struggled to surpass an 85% threshold across these metrics, the proposed model achieved an accuracy of 96.30%. Furthermore, the precision (93.89%), recall (93.55%), and F1-score (93.56%) charts illustrate that our dual-branch approach effectively balances the detection of true positives while minimizing false alarms. This comparison demonstrates that a low-parameter count system can surpass heavy single-model baselines across all performance categories when spatial and texture features are fused.

F. Explainable AI (XAI) Visualization

The Gradient-weighted Class Activation Mapping method (Grad-CAM) has been employed in the Deep Learning approach in order to better understand the "black box" nature of the technology and to provide diagnostic confidence to aquaculture farmers [14]. We applied Grad-CAM to the final convolutional layer ('top_activation') of the EfficientNetV2-B0 Texture branch, in order to visualize the locations of the input that had the greatest effect on the model's decision-making.

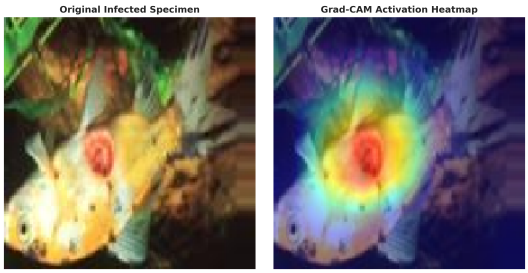


Fig. 4. Grad-CAM visualization on the EfficientNetV2-B0 branch. The heatmap's high-activation regions (red/yellow) confirm the model accurately targets localized pathological features rather than background environmental noise.

Figure 4 presents a jet colormap to visualize where the model was activated, with red and yellow being the most highly activated points. Using the example of the Fish Bacterial Red Disease sample, areas of activation corresponded well with recent localized haemorrhagic lesions (mainly on the body and head) on the fish being analysed. The absence

of activation points in the background image confirms that the model effectively separates actual pathological characteristics from environmental conditions, such as water turbidity and surface reflections. This qualitative evidence provides additional confidence that the model has a high level of accuracy, and the features extracted from the data will be medically applicable to the models in the future.

V. Conclusion and Future Work

As global aquaculture continues to grow quickly, there is an urgent need for rapid, dependable and minimal hardware diagnostic tools to enhance aquaculture management systems. The heterogeneous ensemble architecture proposed in this research bridges the local/global feature divide with an efficient hybrid EfficientNetV2-B1 branch to provide global context and EfficientNetV2-B0 branch for local texture. The soft-voting ensemble results reached a state-of-the-art level by achieving 96.30% accuracy on 11 types of increasingly complex aquatic diseases, and had zero false positives returned as results for healthy shrimp, plus very low false positive rates on healthy fish so that farmers can avoid using unnecessary and costly chemicals.

The EfficientNetV2 architecture provided a 70% reduction in computational parameters relative to earlier used models (e.g. VGG-16) thus allowing for the use of standard mobile edge devices in remote aquaculture applications. Finally, Grad-CAM visualizations for diagnosing and verifying model performance show that the model accurately explored relevant pathological features and not environmental artifacts.

Future research plans include examining the possibility of Knowledge Distillation as a means to compress the model further to allow for real-time inference from low-power IoT cameras. Future work will also focus on incorporating additional modalities including real-time water quality; and integrating/modeling the impact of various water quality components on aquatic disease diagnostics.

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